

Unit & Integration Testing

Chinna pieces-la irundhu serthu test pannradhu

Unit testing = oru function / class thani-a test. **Integration testing** = pala modules serndhu sariya work aagudhaa nu test.

Unit Testing

- Developer than ezhudhuvaanga.
- Fast-a run aaganum, database / network thevai illa (mock use pannunga).
- Oru test = oru behaviour.
- Naming: test_login_with_wrong_password_fails.

AAA Pattern

Step	Meaning
Arrange	Test-ku thevaiyaana data, setup.
Act	Function-a call pannunga.
Assert	Result expected-a irukka nu check.

Integration Approaches

Approach	Meaning
Big Bang	Ellaa modules-um serthu oree neram-la test.
Top-down	Mela irundhu keezha, stubs use pannunga.
Bottom-up	Keezha irundhu mela, drivers use pannunga.
Sandwich	Rendum mix.

Yenna Test Pannanum (Integration)

- Frontend-Backend API calls.
- Backend-Database read / write.
- Third-party services (payment, email).
- Data format sariya pass aagudhaa.

Tools

Language	Unit test tool
Python	unittest, PyTest
Java	JUnit
JavaScript	Jest, Mocha
API testing	Postman

Tip: Bug kandupidicha, adhukku oru test ezhudhi vaainga, thirumba varaama irukkum.